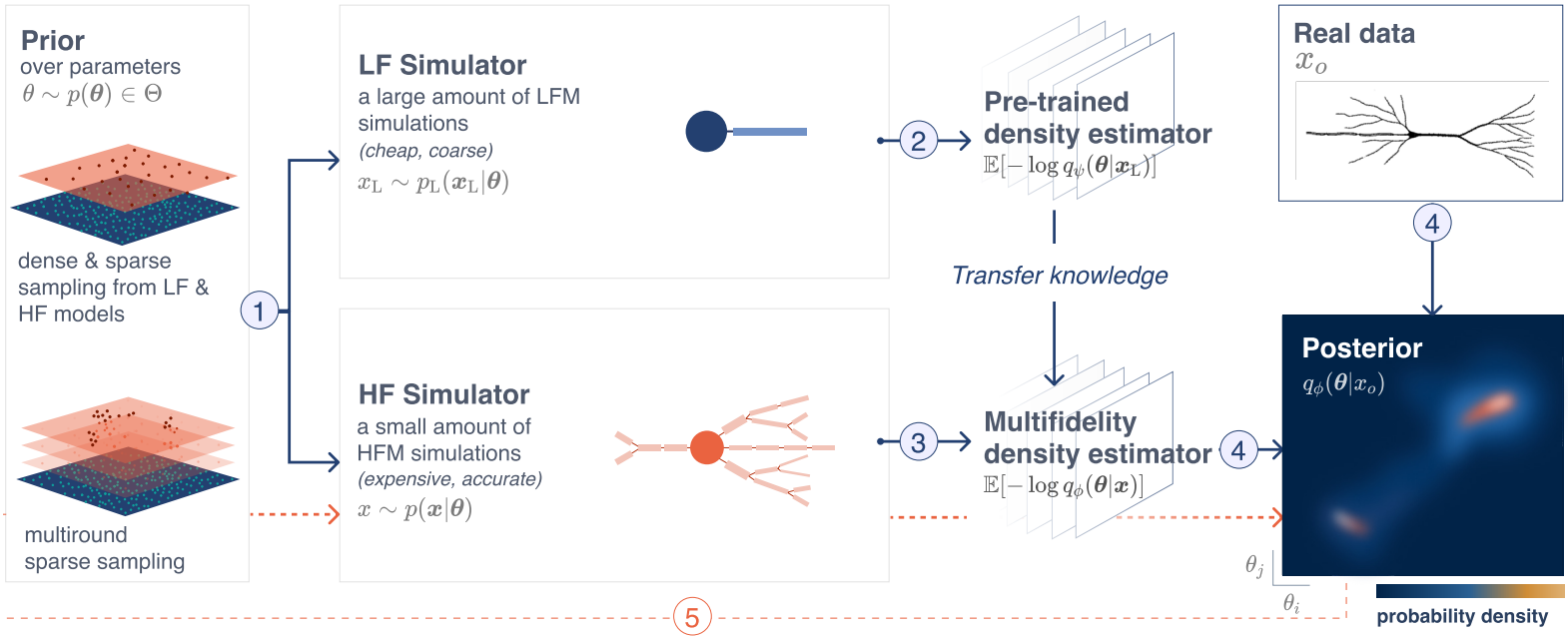
Prior
over parameters
$\theta \sim p(\boldsymbol{\theta}) \in \Theta$
dense & sparse sampling from LF & HF models
multiround sparse sampling
1
LF Simulator
a large amount of LFM simulations
(cheap, coarse)
$x_{\mathrm{L}} \sim p_{\mathrm{L}}(\boldsymbol{x}_{\mathrm{L}}|\boldsymbol{\theta})$
HF Simulator
a small amount of HFM simulations
(expensive, accurate)
$x \sim p(\boldsymbol{x}|\boldsymbol{\theta})$
2
Pre-trained density estimator
$\mathbb{E}[-\log q_\psi(\boldsymbol{\theta}|\boldsymbol{x}_{\mathrm{L}})]$
Transfer knowledge
3
Multifidelity density estimator
$\mathbb{E}[-\log q_\phi(\boldsymbol{\theta}|\boldsymbol{x})]$
4
Real data
x_o
4
Posterior
$q_\phi(\boldsymbol{\theta}|x_o)$
θ_j
θ_i
probability density
5